



Toegepaste Micro-economie: Public economics

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Week 2:
Externalities

Lecture 2.5

Today: Applications of our theory / framework to

- Education
 - Government intervention in education: why?
→ applying the framework
 - Public provision vs public production → book
- ‘Tragedy of the commons’
 - Specific type of negative externalities:
Common-pool problem

Education

Some figures from the OECD (2024) report
‘Education at a Glance’:
Expenditures on education, 2021

Country	Total (% GDP)	% of which public
NL	5.3	80
US	5.8	68
UK	6.2	64
GER	4.6	86
FRA	5.3	86
ITA	4.0	86

Education

Some figures from the OECD (2024):

Exp. per student in tertiary-type A education, 2021

Country	Exp / student (\$)
NL	23.864
US	36.274
UK	33.574
FRA	20.444
ITA	13.717
NOR	26.299
SWE	27.765

That is about
8 times what
students paid!

Education

Some figures from the OECD (2024):

Tuition fee in tertiary-type A education, 2022/2023

Country	Exp / student (\$)	Tuition fee (\$)
NL	23.864	3.041
US	36.274	9.596 – 34.041
UK	33.574	13.135
FRA	20.444	252 - 360
ITA	13.717	2.570 – 8.132
NOR	26.299	0
SWE	27.765	0

Education

From a private perspective (higher) education is (partially) an investment

- Cost: tuition fee; foregone earnings
- Benefit: great lectures (consumption), higher income / productivity afterwards, more interesting jobs, knowledge intrinsically valuable, better health(?),

Education

Why so much government intervention in education?

→ Use our framework!

- Market failures?
- Distributional issues?
- How would people respond?

Here, we focus on higher education (most issues are also relevant for primary & secondary education)

Education

Is education a public good?

- Rival: More pupils or students imply higher cost of providing education
- Excludable: exclusion possible (if allowed for by law)

→ Education is a private good!

Education

External effects of education?

→ Yes, large positive externalities!

- Social benefit > private benefit

- Income tax: “Fiscal externality”

- Spillover effects of knowledge & skills

- Informed participation in collective decision-making

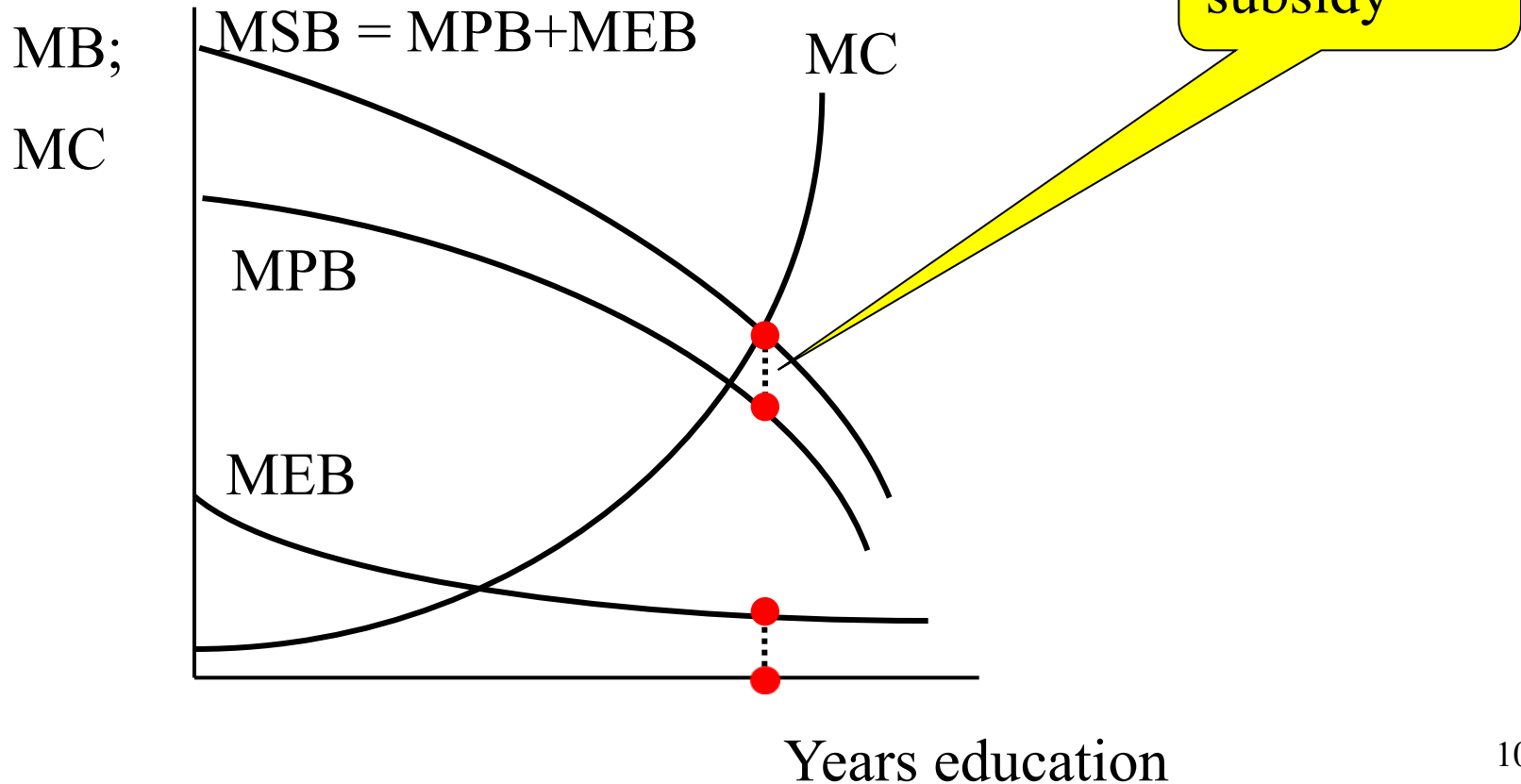
- More trust, lower health care cost, more pro-social behavior (?)

- ...

Offers rationale for subsidizing education

Education

Without subsidy, number of students / level of education inefficiently low



Education

Related: Access to higher education and (high) tuition

- Wealthy students can pay out-of-pocket; poor students have to find a loan.
- Loans for human capital investment risky for banks (no collateral) → high interest rates

(Problem enhanced by asymmetric info; see Lecture 3; implying market failure on market for loans)

- May reduce enrollment by poor students for whom enrolling is efficient
- and leads to (too?) large, costly student debt

Both outcomes are inefficient

Education

Related: Access to higher education and (high) tuition

But: is a generic reduction of tuition fees the proper response to this inefficiency?

Different (better?) solution: government loans

- Reasonable interest rates; lenient repayment rules, etc.
- ‘sociaal leenstelsel’

Education

Asymmetric information (main topic of week 3)?

→ Not the main reason for public provision

One issue: observing quality of education may be difficult for pupils / students / parents

→ Gives schools room to engage in moral hazard

Government regulation / inspection may help to preserve quality / shut down bad schools.

But: can governments determine quality?

Education

Distributional issues / fairness, equality; paternalism

1. Public provision of education reduces intergenerational transmission of differences in income and wealth.
 - Equal opportunities ‘ex ante’, independent of background and income family
 - Education as a means to self-fulfillment (Amartya Sen)
 - Both related to ‘Commodity egalitarianism’
2. Within a cohort, trade-off between efficiency (more investment in high-ability students) and distribution (more investment in low-ability students)
 - That distributional concerns warrant lower university tuition fees is not obvious

Education

How will people respond to higher tuition fees?

1. Decision to enter higher education: see above.
2. Stronger incentive to graduate in nominal duration
 - May lead to less procrastination....
 - ... but also to fewer valuable(?) extracurricular activities
 - ‘langstudeerdersboete’
3. More working next to study. May reduce time spent on studying
4. ...

Public provision of education: to conclude

Application of our framework to education shows:

- Key market failure is (positive) externalities
- Magnitude of externalities largely determines optimal subsidy (or tuition fee): important empirical question!
- Distributional issues are less clear-cut

Many follow-up questions:

- Differentiate by field of studies? Length of studies?
- Students doing multiple studies?
- Distributional issues in an international context
- Should governments regulate content of curriculum?

Public provision versus public production

Common way of public provision: → Public schools

Why not rely more on private production?

- Vouchers for parents / students
- Schools decide about quality and cost (competition!)
- If cost > value voucher, parent / student pays out-of-pocket

Concerns:

- Inequality (rich kids receiving better education)
 - commodity egalitarianism?
- Parents may be unable to judge quality
- Positive peer-effects may be reduced (?).

Concern with public education: **Crowding-out**

→ See book chapter 9.

Externalities: special case

What is common to these issues?:

- Dwindling populations of fish
- Traffic jams
- Strongly reduced flow in China's Yangtze river
- Global warming
- Overcrowded beaches, parks, shopping malls, etc.
- Slow wifi in public places

→ All (partially) caused by *common-pool problem*

Common-pool problem

Common-pool (or –resource) problem can arise when resources are Rival and Non-excluded

- Where ‘non-excluded’ is often caused by insufficiently well-defined and enforced property rights

Rival implies that the marginal cost of more producers / consumers is positive

Non-excluded implies that these cost are not taken into account by individual producers / consumers

Result: **Tragedy of the commons**

Common-pool problem

Tragedy of the commons:

Inefficiencies that arise because individuals do not take into account that their usage of a shared resource affects the costs and/or benefits of other users

Common-pool problem: example

Ork is a fishing community with N inhabitants who only care about money

Each day, each inhabitant of Ork can:

1. Undertake some paid work at wage w , or
2. Rent a boat at cost c , and go fishing.
 - The amount of fish that a fisher catches, F , decreases in the number of fishing-boats B on the lake
→ $F(B)$, where $F'(B) < 0$.
 - Assume: Ork has no market power on fish market
Market price is 1 per fish.

Consequences:

- Revenue from fishing is $F(B)$
- All efficiency effects on producer's side, no effects on consumer surplus

Common-pool problem: example

Market equilibrium number of boats B^P is given by

$$F(B^P) - c - w = 0$$

→ Income from fishing equals outside wage

Socially efficient number of boats B^S :

→ Maximise total income: $\text{Max}_B B[F(B) - c] + (N - B)w$

Which yields first-order condition for B^S :

$$F(B^S) - c - w + B^S F'(B^S) = 0$$

As $B^S F'(B^S) < 0$, we have that $B^P > B^S$

Common-pool problem: example

Why is $B^P > B^S$?

Because in deciding between working and fishing, individuals do not take into account that the presence of their boat on the lake reduce the quantity of fish caught by others.

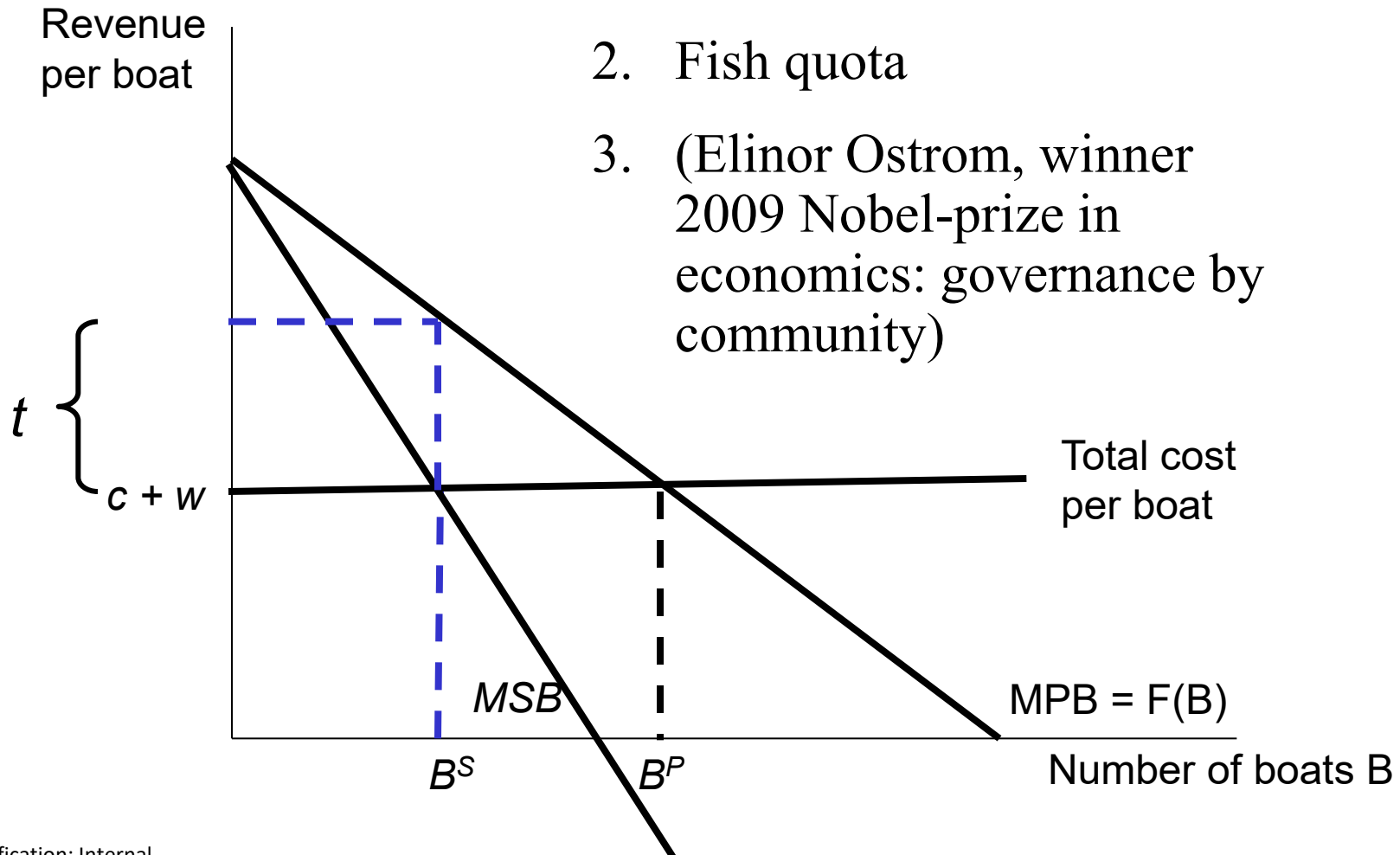
→ This is a negative externality!

Common-pool problem: example

In a figure:

How to get to B^S ?

1. Pigouvian tax per boat $t = -B^S F'(B^S)$
2. Fish quota
3. (Elinor Ostrom, winner 2009 Nobel-prize in economics: governance by community)



Common-pool problem

Tragedy of the commons arises when individuals do not take the effect of their actions on others into account.

Elinor Ostrom (winner 2009 Nobel-prize Economics) has shown that small communities often set up elaborate governance schemes to overcome the common pool problem, and can be successful
→ Coase Theorem (often with implicit contracts)

Concluding remarks

Many policy issues relate to externalities

→ Many examples of choices that affect others, outside the market mechanism

Moreover, some institutional / policy choices have created or enhanced externalities:

- Tax system and poverty alleviation

→ Your choice to invest in productivity-increasing education affects my tax rate

- Collective (health) insurance:

→ My decision not to sport or exercise may affect your insurance premium

Big issues, but also small → read Frank & Cartwright's chapter on externalities (ch 16 (18) in old (new) edition)

Looking ahead

Markets under asymmetric information

- Moral hazard
- Adverse selection
- Application to insurance

- Why millions of people in the US used to have no health insurance
- Why there is no privately provided poverty insurance
- Why the average time-to-graduation of students in NL is probably inefficiently low